THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 02

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The mirror image of the point (5, −2) in the x-axis is:

(a) (−5, −2) (b) (5, 2) (c) (2, −5) (d) (−5, 2)

Ans: (b) Reflection in the x-axis changes only the sign of the y-coordinate.

2. The midpoint of the segment joining A(−4, 7) and B(6, −3) is:

(a) (−5, 5) (b) (1, 2) (c) (1, 5) (d) (5, 2)

Ans: (b) ((−4 + 6)/2, (7 − 3)/2) = (1, 2).

3. If (k, 2) is a solution of the equation 3x + 4y = 20, then the value of k is:

(a) 5 (b) 8 (c) 4 (d) 2

Ans: (c) 3k + 8 = 20 ⇒ 3k = 12 ⇒ k = 4.

4. Which of the following has a terminating decimal expansion?

(a) $\frac{5}{6}$ (b) $\frac{4}{15}$ (c) $\frac{9}{14}$ (d) $\frac{7}{40}$

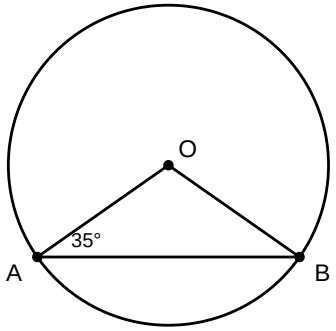
Ans: (d) $40 = 2^3 \times 5$ has only the prime factors 2 and 5, so 7/40 = 0.175 terminates.

5. The value of $51^2 - 49^2$ is:

(a) 200 (b) 2 (c) 2500 (d) 100

Ans: (a) $a^2 - b^2 = (a + b)(a - b) = (100)(2) = 200$.

6. In the figure, O is the centre of the circle and ∠OAB = 35°. Then ∠AOB is equal to:



(a) 35° (b) 70° (c) 145° (d) 110°

Ans: (d) OA = OB (radii) ⇒ ∠OBA = 35°; ∠AOB = 180° − 35° − 35° = 110°.

7. A chord of length 16 cm is drawn in a circle of diameter 20 cm. Its distance from the centre is:

(a) 12 cm (b) 8 cm (c) 2 cm (d) 6 cm

Ans: (d) r = 10 cm; half chord = 8 cm; $d^2 = 100 - 64 = 36$ ⇒ d = 6 cm.

8. The area of an equilateral triangle of side 10 cm, obtained from Heron's formula, is:

(a) $25\sqrt{3}$ cm² (b) $100\sqrt{3}$ cm² (c) $50\sqrt{3}$ cm² (d) 25 cm²

Ans: (a) s = 15; area = $\sqrt{(15 \times 5 \times 5 \times 5)} = 25\sqrt{3}$ cm².

9. A bag contains 5 red, 3 green and 2 blue balls. A ball is drawn at random. The probability that it is not red is:

(a) $\frac{1}{5}$ (b) $\frac{1}{2}$ (c) $\frac{2}{5}$ (d) $\frac{3}{10}$

Ans: (b) Not red = 3 + 2 = 5 balls out of 10 ⇒ 1/2.

10. The n-th term of a sequence is $t_n = n(n - 2)$. Then t_7 is:

(a) 14 (b) 35 (c) 21 (d) 49

Ans: (b) $t_7 = 7 \times 5 = 35$.

11. In which of the following are BOTH the proposition and its converse true?

(a) If $x = y$, then $x^2 = y^2$.

(b) If a number is a multiple of 8, then it is a multiple of 4.

(c) If two triangles have the same area, then they are congruent.

(d) If n is a perfect square, then n has an odd number of factors.

Ans: (d) A positive integer has an odd number of factors exactly when it is a perfect square. In the other three the converse (or the proposition) is false.

12. A counterexample to the claim '$n^2 + n + 11$ is prime for every positive integer n' is:

(a) n = 10 (b) n = 1 (c) n = 3 (d) n = 2

Ans: (a) For n = 10: 100 + 10 + 11 = 121 = 11², which is not prime. (n = 1, 2, 3 give 13, 17, 23.)

13. Jaspreet cycled for 5 days in each of three weeks. Her weekly averages were 12, 18 and 15 minutes. Her average for the 15 days is:

(a) 15 minutes (b) 45 minutes (c) 14 minutes (d) 16 minutes

Ans: (a) All weeks have the same number of days, so the simple average (12 + 18 + 15)/3 = 15 minutes is correct.

14. A cylindrical drum has radius 7 cm and height 10 cm. Its volume ($\pi = 22/7$) is:

(a) 3080 cm³ (b) 770 cm³ (c) 1540 cm³ (d) 440 cm³

Ans: (c) $V = \pi r^2 h = 22/7 \times 49 \times 10 = 1540$ cm³ (440 cm² would be the curved surface area).

15. In the digit-by-digit addition algorithm the carry can never exceed 1 because the largest possible column total is:

(a) 9 + 9 = 18 only (b) 10 + 10 = 20

(c) 9 + 9 + 9 = 27 (d) 9 + 9 + 1 = 19

Ans: (d) Each column adds two digits (at most 9 each) and a carry of at most 1, giving at most 19, i.e. a units digit and carry 1.

16. The diagonals AC and BD of a parallelogram ABCD meet at O. If AC = 14 cm and BD = 10 cm, then OA + OB is:

(a) 10 cm (b) 12 cm (c) 7 cm (d) 24 cm

Ans: (b) Diagonals bisect each other: OA = 7 cm, OB = 5 cm.

17. The sides of a triangle are 10 cm, 14 cm and 18 cm. The perimeter of the triangle formed by joining the midpoints of its sides is:

(a) 21 cm (b) 14 cm (c) 28 cm (d) 42 cm

Ans: (a) Each side of the new triangle is half of a side of the original (Midpoint Theorem), so the perimeter is 42 ÷ 2 = 21 cm.

18. The 10th term of the arithmetic progression 4, 9, 14, … is:

(a) 44 (b) 54 (c) 49 (d) 45

> **Ans: (c)** a = 4, d = 5: $t_{10} = 4 + 9 \times 5 = 49$.

19. **Assertion (A):** 0.333… (3 repeating) is a rational number.
Reason (R): Every repeating decimal can be written in the form p/q, where p and q are integers and q ≠ 0.

> **Ans: (a)** Both true; R explains A since 0.333… = 1/3.

20. **Assertion (A):** If a proposition is true, then its converse is always true.
Reason (R): The proposition 'If it rains, then the road is wet' is true, but the road can be wet without rain.

> **Ans: (d)** R is a true counterexample to A, so A is false.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. An e-rickshaw charges ₹30 for the first 2 km and ₹9 for every additional km. Find the fare for a journey of 8 km.

> Extra distance = 8 − 2 = 6 km (½)
> Extra charge = 6 × 9 = ₹54 (1)
> Fare = 30 + 54 = **₹84** (½)

22. Without actual division, decide whether $\frac{7}{125}$ has a terminating decimal expansion. If so, state the number of decimal places.

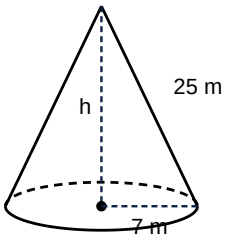
> 125 = 5^3 has only the prime factor 5 ⇒ terminates (1). Number of decimal places = **3** (0.056) (1)

O R

Find two rational numbers between $\frac{3}{8}$ and $\frac{1}{2}$.

> 3/8 = 12/32 and 1/2 = 16/32 (1). Two numbers: **13/32 and 14/32** (any two correct numbers) (1)

23. A conical tent has base radius 7 m and slant height 25 m. Find its curved surface area and its height (π = 22/7).

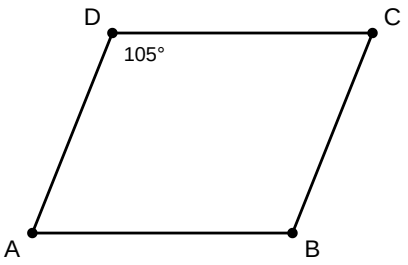


> CSA = πrl = 22/7 × 7 × 25 = **550 m²** (1)
> h = √(25² − 7²) = √576 = **24 m** (1)

24. A dairy mixes 400 litres of milk containing 3% fat with 100 litres of milk containing 8% fat. Find the percentage of fat in the mixture.

> Fat = 400 × 0.03 + 100 × 0.08 = 12 + 8 = 20 litres (1)
> Percentage = 20 ÷ 500 × 100 = **4%** (1)

25. In the parallelogram ABCD shown, ∠ADC = 105°. Find ∠DAB and ∠ABC, giving reasons.



Adjacent angles of a parallelogram are supplementary: **∠DAB = 180° − 105° = 75°** (1)
Opposite angles are equal: **∠ABC = ∠ADC = 105°** (1)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express 2.3636… (36 repeating) in the form p/q in lowest terms.

x = 2.3636…; 100x = 236.3636… (1)
100x − x: 99x = 234 (1)
x = 234/99 = **26/11** (1)

27. A survey of 200 families recorded the number of children in each family:

Children	0	1	2	3
Families	30	80	70	20

A family is chosen at random. Find the probability that it has (i) no child, (ii) exactly one child, (iii) at least two children.

Total = 200. (i) 30/200 = **3/20** (1) (ii) 80/200 = **2/5** (1) (iii) (70 + 20)/200 = **9/20** (1)

OR

A bag contains 7 red, 5 white and 8 black balls. One ball is drawn at random. Find the probability that it is (i) red, (ii) not white, (iii) white or black.

Total = 20. (i) **7/20** (1) (ii) 15/20 = **3/4** (1) (iii) (5 + 8)/20 = **13/20** (1)

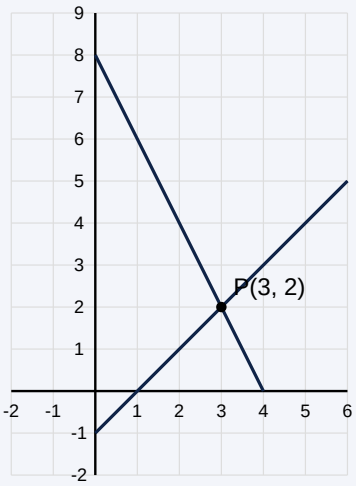
28. Draw the graphs of the equations 2x + y = 8 and x − y = 1 on the same grid and use them to find the solution of the pair.

Table for 2x + y = 8:

x	0	4
y	8	0

Table for x − y = 1:

x	0	1
y	-1	0



(1) Draw both lines (1) The lines meet at **(3, 2)**, so x = 3, y = 2 (1)

29. Write the converse of the proposition ‘If a quadrilateral is a rhombus, then its diagonals are perpendicular.’ Decide whether the converse is true and give a counterexample if it is not.

Converse: **If the diagonals of a quadrilateral are perpendicular, then it is a rhombus.** (1)
False (1). Counterexample: a kite with unequal adjacent sides has perpendicular diagonals but is not a rhombus (1).

30. A chord of length 12 cm is drawn in a circle of radius 10 cm. Find (i) its distance from the centre; (ii) the length of another chord of the same circle at a distance of 6 cm from the centre.

(i) Half chord = 6: $d^2 = 100 - 36 = 64 \Rightarrow$ **d = 8 cm** (1½)
(ii) Half chord^2 = 100 − 36 = 64 ⇒ half chord 8 ⇒ chord = **16 cm** (1½)

O R

Using Heron’s formula, find the area of a triangle with sides 15 cm, 36 cm and 39 cm.

s = (15 + 36 + 39)/2 = 45 cm (1)
Area = √[45 × 30 × 9 × 6] (1) = √72900 = **270 cm²** (1)

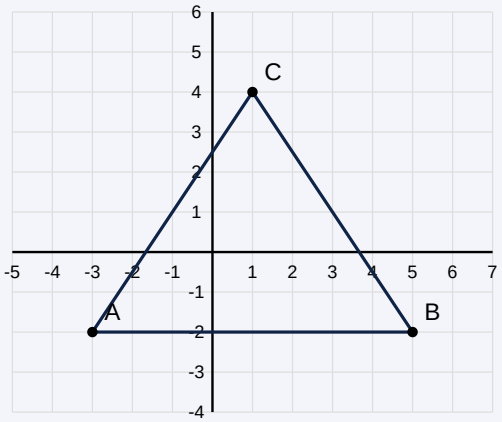
31. Write an algorithm to find the lcm of two numbers m and n by testing the numbers max(m, n), max(m, n) + 1, … in turn. Execute it for m = 4 and n = 6.

Algorithm: (1) Let L = max(m, n). (2) If both m and n divide L, report L and stop. (3) Otherwise increase L by 1 and go back to step 2. (1½)
For 4 and 6: L = 6, 7, 8, 9, 10, 11 fail; **L = 12** is divisible by both ⇒ lcm = 12 (1½)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. The three corners of a triangular flag are marked on a grid as A(−3, −2), B(5, −2) and C(1, 4).
(a) Plot the points and join them. (2 marks)
(b) Find the lengths AB, BC and CA. (1½ marks)
(c) Name the type of triangle on the basis of its sides and find its area. (1½ marks)



(a) Plotting 3 points (1½), joining (½): (b) AB = 8; BC = $\sqrt{(4^2 + 6^2)} = \sqrt{52}$; CA = $\sqrt{(4^2 + 6^2)} = \sqrt{52}$ (1½)
(c) BC = CA ⇒ **isosceles triangle** (½). Base AB = 8 and the height from C = 4 − (−2) = 6, so area = ½ × 8 × 6 = **24 square units** (1)

33. How many multiples of 8 lie between 200 and 500? Find their sum. (2½ + 2½)

Multiples: 208, 216, …, 496 — an AP with a = 208, d = 8, l = 496 (1)
n = (496 − 208)/8 + 1 = **37** (1½)
Sum = 37/2 × (208 + 496) = 37 × 352 = **13024** (2½)

O R

The 5th term of an AP is 17 and its 12th term is 38. Find the AP and the sum of its first 20 terms. (2 + 3)

a + 4d = 17 and a + 11d = 38 ⇒ 7d = 21 ⇒ d = 3, a = 5 (1½): AP: 5, 8, 11, … (½)
S_{20} = 20/2 × [2(5) + 19(3)] (1½) = 10 × 67 = **670** (1½)

34. Prove that if the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Let diagonals AC and BD of quadrilateral ABCD meet at E with EA = EC and EB = ED. (1)
In △AEB and △CED: EA = EC, EB = ED, ∠AEB = ∠CED (vertically opposite) ⇒ △AEB ≅ △CED (SAS) ⇒ ∠ABE = ∠CDE (1½)
These are alternate angles for AB and DC with transversal BD, so AB ∥ DC (½)
Similarly △AED ≅ △CEB (SAS) ⇒ ∠ADE = ∠CBE ⇒ AD ∥ BC (1½)
Both pairs of opposite sides are parallel ⇒ **ABCD is a parallelogram**. (½)

O R

State and prove the converse of the Baudhāyana–Pythagoras theorem: if $a^2 + b^2 = c^2$ for the sides of a triangle, then it is right-angled.

Statement: in △ABC with BC = a, CA = b, AB = c, if $a^2 + b^2 = c^2$, then ∠C = 90°. (1)
Construct a right triangle XYZ with ∠Z = 90°, YZ = a and XZ = b (1)
By the Pythagoras theorem $XY^2 = a^2 + b^2 = c^2$ ⇒ XY = c (1½)
Now △ABC and △XYZ have BC = YZ, CA = ZX and AB = XY ⇒ △ABC ≅ △XYZ (SSS) (1)
Hence ∠C = ∠Z = 90°: **△ABC is right-angled**. (½)

35. The area of a rectangular carpet is $(8x^2 + 14x + 3)$ m² and its breadth is (2x + 3) m. Factorise the expression to find its length. If x = 3, find the perimeter of the carpet.

$8x^2 + 14x + 3 = 8x^2 + 2x + 12x + 3$ (1½) = 2x(4x + 1) + 3(4x + 1) = (2x + 3)(4x + 1) (1)
Length = **(4x + 1) m** (1)
For x = 3: breadth 9 m, length 13 m ⇒ perimeter = 2(9 + 13) = **44 m** (1½)

O R

The sum of the length and breadth of a rectangular notice-board is 17 dm and its area is 60 dm². Form a quadratic expression in the breadth x, factorise it and find the length and breadth.

Length = (17 − x). Area: x(17 − x) = 60 ⇒ $x^2 - 17x + 60 = 0$ (1½)
Factorise: (x − 5)(x − 12) = 0 (1½)
x = 5 or 12 (1). The board is **12 dm by 5 dm** (1)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A drone delivery company maps its city on a coordinate grid (1 unit = 100 m). The warehouse W is at the origin, the shop S is at (6, 8), the school T is at (−6, 8) and a house H is at (−3, −4).

(i) Find the distance of the shop S from the warehouse W (in grid units).

WS = $\sqrt{(6^2 + 8^2)}$ = **10 units**

(ii) Find the coordinates of the midpoint of the segment ST.

((6 − 6)/2, (8 + 8)/2) = **(0, 8)**

(iii) Show that the warehouse is at the same distance from S and T.

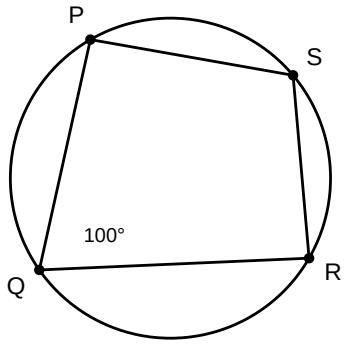
WS = 10 (½); WT = $\sqrt{((-6)^2 + 8^2)}$ = 10 (1) ⇒ WS = WT (½)

OR

(iii) A drone flies directly from H to S. Find the distance it covers in metres.

HS = $\sqrt{[(6 + 3)^2 + (8 + 4)^2]}$ = $\sqrt{(81 + 144)}$ = 15 units (1) ⇒ 15 × 100 = **1500 m** (1)

37. Four floodlights P, Q, R and S are fixed on the boundary of a circular ground (in that order). It is known that ∠PQR = 100° and ∠QPS = 85°.



(i) Find ∠PSR.

Opposite angles of a cyclic quadrilateral are supplementary: ∠PSR = 180° − 100° = **80°**

(ii) Find ∠QRS.

∠QRS = 180° − ∠QPS = 180° − 85° = **95°**

(iii) If ∠QPR = 40°, find ∠QSR and state the reason.

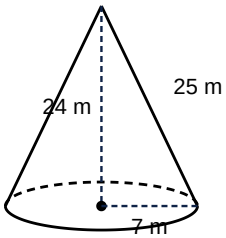
Angles in the same segment (on chord QR) are equal (1) ⇒ **∠QSR = 40°** (1)

OR

(iii) The side PS is produced to a point T. Find the exterior angle ∠RST and state the property used.

Exterior angle of a cyclic quadrilateral equals the interior opposite angle (1) ⇒ ∠RST = ∠PQR = **100°** (1). (Also ∠RST = 180° − 80° = 100°.)

38. A grain merchant stores grain in a conical heap of base radius 7 m and height 24 m (π = 22/7).



(i) Find the slant height of the heap.

> $l = \sqrt{(24^2 + 7^2)} =$ **25 m**

(ii) Find the volume of grain in the heap.

> $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 22/7 \times 49 \times 24 =$ **1232 m³**

(iii) Find the area of canvas needed to cover the heap.

> CSA = πrl = 22/7 × 7 × 25 (1) = **550 m²** (1)

O R

(iii) The grain is poured into a cylindrical bin of the same base radius. Find the height of grain in the bin.

> $\pi r^2 h = 1232 \Rightarrow 154h = 1232$ (1) $\Rightarrow$ **h = 8 m** (1)